

Yogesh Gandhi

PhD in Aerospace Engineering

Computational Mechanics · Finite Element Simulation · Design Optimization

Tokyo, Japan · gandhi.k.yogesh@gmail.com

[gandhiyudheer.github.io](https://github.com/gandhiyudheer) · github.com/gandhiyudheer · ORCID 0000-0002-8330-8515

Computational scientist and mechanical/aerospace engineer with a PhD and seven years of research and industrial experience across computational mechanics, finite element simulation, and engineering design optimization. Work spans aerospace, automotive, and advanced manufacturing — from linear and nonlinear structural analysis to building the numerical infrastructure and software behind automated engineering design. First-author in *CMAME*; production work shipped on parts for Ferrari and the winning Sasol 2024 solar vehicle. Currently developing the simulation engine behind automated engineering design at Braid Technologies, Tokyo.

Technical skills

Languages: Python · MATLAB · Fortran · Bash

Numerics & FE: FEM/FEA (linear & nonlinear, implicit & explicit) · structural & contact mechanics · buckling & stability · dynamics · multiscale modeling

Optimization: design & topology optimization (SIMP, geometry projection, CAD-based) · gradient-based methods · manufacturing & design constraints

Solvers & tools: LS-DYNA · Kratos Multiphysics · MFEM · Abaqus · ANSYS · SolidWorks · GMSH

Systems: Linux · Git · LaTeX · scientific computing & data analysis (NumPy, SciPy, JAX)

Spoken: English (C1) · Italian (B2) · Japanese (learning)

Experience

Senior Researcher — *Braid Technologies K.K., Tokyo, Japan*

Aug 2025 – Present

- Develop the mathematical models and FE building blocks that power Braid's automated design engine, serving Tier-1 automotive and manufacturing clients.
- Built and integrated nonlinear-static, contact, and constraint modules; debugged convergence in 3D implicit buckling with simultaneous geometric, material, and contact nonlinearities.
- Implemented new element formulations (thin shells, corotational kinematics) and orchestrate multi-solver strategies across LS-DYNA, Kratos Multiphysics, and MFEM.
- Maintain mesh-processing pipelines spanning TRI3/TRI6, hex, and IGA-ready geometries.

Postdoctoral Researcher — *University of Bologna, Forlì, Italy*

Jan – Jun 2024

- Translated PhD-stage methods into production CFRP design workflows in the ASCENT and OptiFiber projects.
- Delivered stress-based topology optimization for 3D-printed wing-mechanism parts of the Ferrari 296GTB.
- Co-led design and additive manufacturing of structural parts for the Emilia V solar vehicle — winner of the Sasol 2024 Cruiser Challenge.

Research Associate — *University of Parma, Parma, Italy*

Apr 2018 – Feb 2020

- Implemented thermo-mechanical UMAT subroutines (Abaqus) for shape memory alloys; derived fiber-layup optimization for bistable laminates.
- Built VCCT and cohesive-element delamination models calibrated against experimental data.

Earlier: Visiting Graduate Researcher, Coventry University, UK (2017) — path-planning (A^* , D^*) and nonlinear control (Particle Swarm Optimization) for autonomous rovers. Research collaborations with TU Delft, Politecnico di Torino, and Università di Padova (2021–2025) — CAD-based topology optimization, Carrera Unified Formulation for laminates, and additive process modeling.

Education

PhD, Aerospace Engineering — University of Bologna, Forlì, Italy

2020 – 2024

Geometry projection method for additively manufactured variable-stiffness continuous fiber-reinforced polymer structures.

MSc, Aerospace Engineering — University of Bologna, Forlì, Italy

2015 – 2017

Selected projects

- **Geometry projection for variable-stiffness composites.** Unified topology-optimization framework for additively manufactured composite laminates with continuous fibers; designs layout, orientation, and stiffness in one formulation. *CMAME*, 2025.
- **Ferrari 296GTB wing mechanism (OptiFiber).** Stress-based design framework for 3D-printed composite parts; topology-optimization methodology and experimental validation.
- **Emilia V solar vehicle.** Co-led structural design and additive manufacturing of components; first place at the Sasol 2024 Cruiser Challenge, South Africa.
- **TO with Carrera Unified Formulation (with POLITO).** Extended topology optimization via CUF for accurate through-thickness modeling of multilayered laminates.

Selected publication

126 citations · h-index 5 · i10-index 4 (Google Scholar, 2026)

Y. Gandhi, A. Aragón, J. Norato, G. Minak. *A geometry projection method for the topology optimization of additively manufactured variable-stiffness composite laminates*. *Computer Methods in Applied Mechanics and Engineering*, 2025. doi:10.1016/j.cma.2024.117663

Full publication list at gandhiyudheer.github.io/#publications and ORCID.

Teaching, mentoring & service

- **PhD co-supervisor**, University of Bologna (2024 – present) — design framework for 3D-printed parts using Emilia V experimental data. **MSc co-supervisor** (2024) — stress-based TO for AM structures.
- **Lecturer:** MSc Mechanical Eng, University of Bologna (2023, 2024) — topology-optimization theory & MATLAB; MSc Automotive Eng, Dallara Academy (2019) — composite failure & delamination in Abaqus.
- **Peer reviewer** for Composite Structures and Additive Manufacturing.